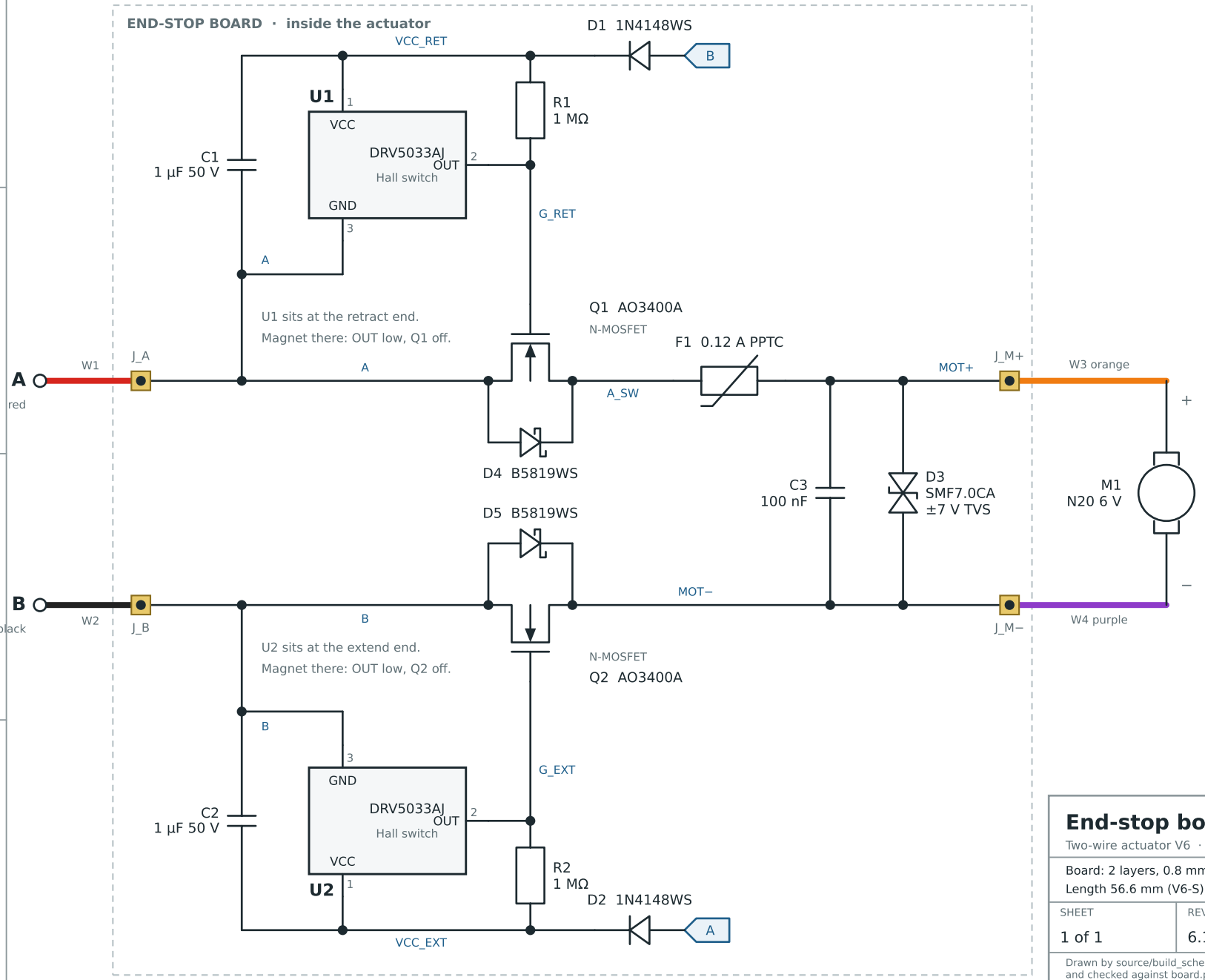
V6 end-stop board

Two wires in. Polarity sets the direction; a Hall switch at each end cuts that direction off, and reversing always works. No microcontroller.



HOW IT WORKS

A positive, B negative: the rod extends.
Current: A → D4 → F1 → M1 → Q2 → B.
At the extend end U2 sees the magnet, pulls Q2's gate low, and the motor stops.

B positive, A negative: the rod retracts.
Current: B → D5 → M1 → F1 → Q1 → A.
U1 stops it at the retract end the same way.

At an end, the opposite polarity still runs: D4 / D5 (and the MOSFET body diodes) carry the current around the switch that is off.

Each Hall switch is fed from the other wire through D1 / D2, so it only runs while its own direction is driven. C1 / C2 hold its supply.

PROTECTION

F1: resettable fuse, 0.12 A hold. A jammed rod trips it in about 1.6 s (simulated); it resets when the power is removed for a few seconds. D3 clamps the motor's switch-off spike; C3 suppresses brush noise.

DRIVE

Any H-bridge or reversing (DPDT) switch.
6 V nominal, 7 V maximum (D3). PWM up to 20 kHz simulated. Stopped at an end, the board draws about 2.2 mA (the Hall switch).

End-stop board · schematic

Two-wire actuator V6 · N20 6 V 150 rpm gearmotor with M3 leadscrew

Board: 2 layers, 0.8 mm FR4, 8 mm wide, all parts on one side.
Length 56.6 mm (V6-S), 76.6 mm (V6-U60) or 101.6 mm (V6-L).

SHEET	REVISION	DATE	V6 BOARDS
1 of 1	6.1	2026-10-28	S · U60 · L

Drawn by source/build_schematic.py; connectivity re-derived from the drawing and checked against board.py: 9 nets, 40 pins, 22 junctions, no differences.